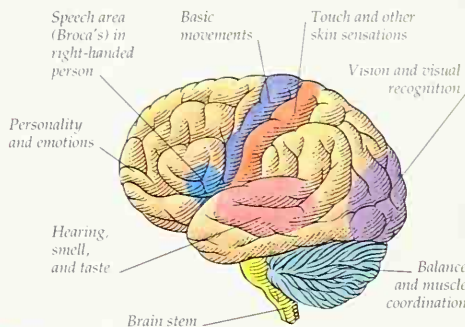
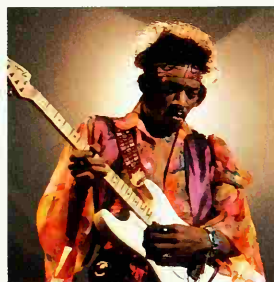


LEFT AND RIGHT

In the spinal cord and base of the brain, nerves cross from left to right. This means the left side of the brain receives sensory signals from, and sends motor signals to, the right side of the body, and vice versa. One side of the brain usually dominates, and this relates to a person's "handedness." The left side dominates in right-handers – it has Broca's area for example – and in left-handed people, Broca's area is generally found on the right cerebral hemisphere. Left-handed people often excel in the visual, musical, and the creative arts, such as extraordinary rock guitarist Jimi Hendrix (1942-1970), who held his right-handed guitar upside down.



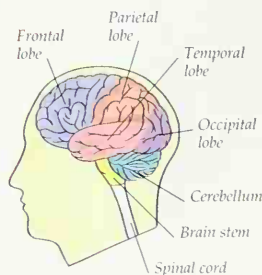
BRAIN CENTERS

Experiments on animals and studies of people with brain diseases have revealed that certain parts of the brain have specific roles. Patches or "centers" of the cortex – the outer layer of the cerebral hemisphere – are the main processing areas for functions such as hearing, touch, speech, sight, and movement.



Right cerebral hemisphere

THE BRAIN FROM ABOVE
Blood that has flowed through the upper brain tissues and veins collects in large, thin-walled spaces known as sinuses of the brain. This model shows the heavily folded structure of the cerebrum, with bulging folds called gyri separated by grooves named sulci. The pattern of folding varies from one person to another. The brain is covered with three membranes called meninges. These are called the dura mater (p. 57), arachnoid, and pia mater.



LOBES OF THE BRAIN

Each cerebral hemisphere of the brain has four main lobes, marked by major grooves. They are named after the bones of the skull (p. 15) around them – the frontal, parietal, temporal, and occipital lobes.